Exercise

MED-EL

Vehicle assembly

The task is to create an assembly to encapsulate different vehicles from a client application. For that purpose, you negotiated with the customer that there needs to be a method `void ShowInformation()` which outputs on the debug line and shows the vehicle information, for example:

```
myCar.ShowInformation();
```

"Driving a car from Toyota."

```
myBike.ShowInformation();
```

"Driving a motorcycle from Honda."

The customer further requested that the framework supports the following vehicles:

- Motorcycles fabricated by Honda
- Motorcycles fabricated by KTM
- Cars fabricated by Toyota
- Cars fabricated by Honda

Additionally, the customer requests that you implement a function for switching the tires of cars, and also show which tires (including its properties) a car uses by calling the `ShowInformation()` method. Summer tires are characterized by pressure and maximum temperature. Winter tires are characterized by pressure, minimum temperature and thickness. Assume all values are floats. Cars should come per default with summer tires, where pressure is set to 2.5 bar, and the maximum temperature to 50°C.

Tire fitting shop assembly

Create another assembly for a multithreaded real-time simulation of an automobile tire fitting shop where the total number of customers and the number of customers whose tires can be changed simultaneously are defined before the simulation starts. The time required to change tires must be a random value between 2 and 5 seconds. Customers should arrive sequentially, and the delay between arrivals should be a random value in the range of 0 to 1 second. The simulation should run until the expected number of customers have been served. The simulation events should be logged to the debug output as they occur in real-time.

The example of a simulation debug output where the total number of customers is 5 and the number of simultaneously served customers is 2:

```
Timestamp (s)   Message
0.2             Customer 1 has arrived. Driving a car from Toyota...
0.2             Customer 1 car tires are being changed and it will take 2.3 seconds.
0.2             Customer 2 has arrived. Driving a car from Honda...
0.2             Customer 2 car tires are being changed and it will take 3.6 seconds.
0.3             Customer 3 has arrived. Driving a car from Toyota...
0.9             Customer 4 has arrived. Driving a car from Toyota...
1.2             Customer 5 has arrived. Driving a car from Honda...
2.5             Customer 1 has left.
2.5             Customer 3 car tires are being changed and it will take 4.4 seconds.
3.9             Customer 2 has left.
3.9             Customer 4 car tires are being changed and it will take 3.4 seconds.
6.9             Customer 3 has left.
6.9             Customer 5 car tires are being changed and it will take 3.5 seconds.
7.3             Customer 4 has left.
10.4            Customer 5 has left.
```

Verify the implementation of the tire fitting shop with 100 customers and 4 mechanics.

Implement the functionality by applying object-oriented principles in C# and create high-quality, production-ready source code. In case you have any questions or need additional information please feel free to contact us.